

Employee Attrition Analysis

Summary Report for HR Leadership

Employees Analysed	1,470	Overall Attrition Rate	16.1%	Highest-Risk Job Role	Sales Rep (~40%)	Model Prediction Score	80% AUC
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Background & Objective

This analysis examined records of 1,470 employees to understand why people leave the organisation and to build a tool that can help HR identify at-risk employees before they resign. Three predictive models were tested; the best one correctly identifies roughly 8 in 10 potential leavers, giving HR teams time to intervene.

Who Is Leaving — and Why?

By Department

Department	Attrition Rate
Sales	~20.5%
Human Resources	~19.0%
Research & Development	~13.8%

By Job Role (Top 3 at Risk)

Job Role	Attrition Rate
Sales Representative	~40%
Laboratory Technician	~24%
Human Resources	~23%

Key Drivers of Attrition

- Overtime** — Employees required to work overtime are significantly more likely to leave — the single strongest signal in the model.
- Compensation** — Lower monthly income is closely associated with higher turnover. Employees who leave earn noticeably less than those who stay.
- Work-Life Balance** — Those rating their work-life balance as poor are disproportionately represented among leavers.
- Job Satisfaction** — Low satisfaction scores in job role and environment are consistent predictors of attrition.
- Career Stage** — Employees early in their tenure or with fewer years at the company leave at higher rates, suggesting onboarding and early engagement are critical.

How the Prediction Model Works

Three analytical approaches were tested on historical employee data. Each was evaluated on its ability to correctly flag employees who actually left. Because identifying leavers early is the priority, the model was assessed on both its overall accuracy and its ability to catch true at-risk cases — not just make safe guesses.

Model	Overall Score	Catch Rate
Logistic Regression	80.4%	Best
Gradient Boosting	77.6%	Moderate
Random Forest	76.5%	Lower

Logistic Regression was selected as the recommended model. It provides the most reliable early warning, minimising the risk of missing employees who are about to leave.

Recommendations

- Introduce workload controls for overtime-heavy roles, particularly in Sales and HR. Sustainable workloads are the most impactful lever available.
- Review compensation benchmarking for Sales Representatives, Laboratory Technicians, and HR staff — the three highest-attrition roles.
- Launch structured check-ins for employees in their first two years of service, when attrition risk is at its highest.
- Deploy the predictive model as a quarterly HR tool. A short list of flagged employees allows managers to have targeted retention conversations early.
- Investigate work-life balance and job satisfaction scores by team. These are leading indicators and can be acted on before employees decide to leave.

Limitations

The model is trained on historical data and reflects past patterns. It does not account for future organisational changes, personal circumstances, or broader market conditions. Results should inform — not replace — managerial judgement.